Topics

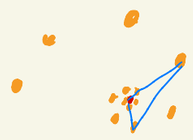
- Completeness of compact spaces, and $\mathbb{R}^k$
- Monotonic convergence
- lim sup and lim inf

Last time:

- Every seq. in compact X has conv. subseq.
- Every bounded seq. in $\mathbb{R}^d$ has conv. subseq.
- Subsequential limits form a closed set



- Convergent $\Rightarrow$ Cauchy



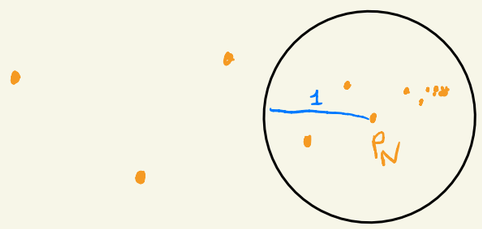
- If Cauchy $\Rightarrow$ Convergent, the space is called complete
- If (p_n) is Cauchy with conv subseq. $p_{n_k} \to q$, then $p_n \to q$.

Thm (3.11b) Compact metric spaces are complete.

proof: Suppose X compact. Let (p_n) be a Cauchy seq in X. (p_n) has a subsequential limit by Thm 3.6, so (p_n) converges. □

Lemma Every Cauchy sequence (p_n) in a metric space X is bounded.

proof: $\exists N$ s.t. $m,n \geq N \Rightarrow d(p_m,p_n) < 1$. Let $R = \max\{1, d(p_1,p_N), d(p_2,p_N), \dots, d(p_{N-1},p_N)\}$ We have $d(p_n,p_N) \leq R$ for all $n \in \mathbb{N}$.



Thm (3.11c) $\mathbb{R}^d$ is complete.

proof: Let (p_n) be a Cauchy seq. in $\mathbb{R}^d$.
Since (p_n) is cauchy, it is bounded. Every bounded sequence in $\mathbb{R}^d$ has a convergent subsequence, so p_n has a convergent subseq. $p_{n_k} \to q$.
Since (p_n) is Cauchy, $p_n \longrightarrow q$.

Aside: How is completeness helpful practically? Showing solutions to equations exist.

Thm (Contraction Mapping Thm) If $\phi : X \longrightarrow X$ is a function such that $d(f(x), f(y)) \leq c\, d(x,y)$ for all $x, y \in X$, and some fixed $c \in (0,1)$, and X is complete then $\phi(x) = x$ has a unique solution.

In a special case, bounded $\Rightarrow$ convergent.

Monotonic sequences

Def A seq. $\{s_n\}$ in $\mathbb{R}$ is

(a) <u>increasing</u> if $s_n \le s_{n+1}$

(b) <u>decreasing</u> if $s_n \ge s_{n+1}$

(c) $\{s_n\}$ is <u>monotonic</u> if it is increasing or decreasing.

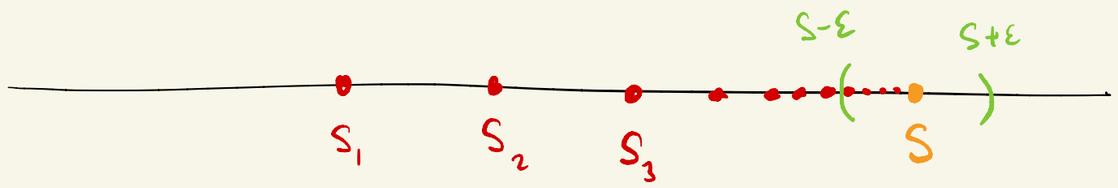
Thm (3.14) Suppose (s_n) monotonic. Then (s_n) converges iff it is bounded.

proof Convergent $\Rightarrow$ bounded already done

Suppose (s_n) increasing: $s_{n+1} \geq s_n$.

Suppose (s_n) bounded. Then there is a least upper bound.

Let $s = \sup\{s_n\}$



Let $\varepsilon > 0$. $\exists N$ s.t. $s - \varepsilon < s_N$, otherwise $s - \varepsilon$ is a smaller upper bound.

For all $n \geq N$, we have $s - \varepsilon < s_N \leq s_n \leq s$ so $s_n \in (s-\varepsilon, s+\varepsilon)$.

(s_n) thus converges to s.

Def: The <u>extended real numbers</u> is the ordered set $\mathbb{R} \cup \{-\infty, \infty\}$ with the order on $\mathbb{R}$ extended s.t. $-\infty < a < \infty$ for all $a \in \mathbb{R}$.

Def Let (s_n) be a sequence in $\mathbb{R}$.

(a) $s_n \to \infty$ means $\forall M \in \mathbb{R}$ $\exists N \in \mathbb{N}$ s.t.

$n \geq N \Rightarrow s_n > M$

(b) $s_n \to -\infty$ means $\forall M \in \mathbb{R}$ $\exists N \in \mathbb{N}$ s.t.

$n \geq N \Rightarrow s_n < M$

Note: we do not say (s_n) is convergent.

Question: If s_n unbounded above, does that imply $s_n \to \infty$?

Def Let (s_n) be a seq in $\mathbb{R}$. Let E be the set of subsequential limits of (s_n) in $\mathbb{R} \cup \{\infty, -\infty\}$.

(a) $\limsup_{n\to\infty} s_n := \sup E$

(b) $\liminf_{n\to\infty} s_n := \inf E$

Examples

1) $\limsup_{n\to\infty} (-1)^n = 1$

$\liminf_{n\to\infty} (-1)^n = -1$

2) $\limsup_{n\to\infty} \sin(n) = 1$

$\liminf_{n\to\infty} \sin(n) = -1$

(not so easy to show)

3) $a_n = \begin{cases} n & n \text{ even} \\ \frac{1}{n} & n \text{ odd} \end{cases}$

$\limsup_{n\to\infty} a_n = \infty$

$\liminf a_n = 0$

